Supplementary Information

Laser-Assisted Nuclear Thermal Propulsion for Cislunar Logistics: A Monte Carlo Power-Closure Assessment

Leon Sandler

Independent Researcher, Northbrook, Illinois, USA. sandler.leon@gmail.com

S1. Thermochemistry

T (K)	p (MPa)	Reference	This model	Difference
2,800	3.0	0.0075	0.0079	5.1%
3,000	0.1	0.0793	0.0836	5.5%
3,500	3.0	0.0542	0.0577	6.5%
4,000	0.1	0.6260	0.6534	4.4%
4,500	3.0	0.3028	0.3264	7.8%
5,500	3.0	0.6982	0.7373	5.6%
6,000	3.0	0.8275	0.8601	3.9%

Table S1. Dissociation degree of hydrogen: this model and Cantera 3.2.0 (h2o2.yaml).

T (K)	p (MPa)	Gibbs minimization	Analytic
3000	0.1	0.083622	0.083622
4500	3.0	0.326411	0.326411
6000	3.0	0.860125	0.860125

Table S2. Dissociation degree from a numerical minimization of the Gibbs energy and from the analytic equilibrium constant (same thermodynamic data).

S2. Power and laser

Thrust (kN)	1300 s	1400 s	1500 s
20	217 (161–350)	267 (196–432)	316 (231–490)
50	543 (404–874)	668 (490–1,079)	790 (578–1,225)
80	868 (646–1,399)	1,069 (783–1,726)	1,263 (925–1,960)
120	1,302 (969–2,099)	1,603 (1,175–2,589)	1,895 (1,388–2,940)
150	1,628 (1,211–2,623)	2,004 (1,469–3,237)	2,369 (1,735–3,675)

Table S3. Emitted laser power (MW; median and 5th–95th percentile) for a given thrust and target specific impulse (start of the burn, full capture).

Aperture (m)	Jitter (µrad)	Range at 0.5	Range at 0.1
5	0.02	27,441	70,314
5	0.05	24,625	63,099
5	0.10	18,883	48,384
5	0.30	8,149	20,880
10	0.02	51,472	131,891
10	0.05	37,777	96,798
10	0.10	23,208	59,620
10	0.30	8,405	21,592
20	0.02	84,219	216,356
20	0.05	46,429	119,276
20	0.10	24,880	63,753
20	0.30	8,470	21,759
40	0.02	110,970	284,343

Aperture (m)	Jitter (µrad)	Range at 0.5	Range at 0.1
40	0.05	49,776	127,544
40	0.10	25,334	65,081
40	0.30	8,492	21,815
100	0.02	124,298	319,318
100	0.05	50,813	130,202
100	0.10	25,464	65,417
100	0.30	8,514	21,815

Table S4. Range (km) at which the captured fraction falls to 0.5 and 0.1 for a 3 m receiver, $M^2 = 1.2$.

Segment	Range (km)	10 m aperture	100 m aperture	Best	Aperture for 0.5 (m)	Receiver radius for 0.5, 10 m aperture (m)
LOI start	1,000	1.000	1.000	1.000	0.5	0.1
TLI end	29,000	0.691	0.880	0.882	6.3	2.3
L1 LLO	60,000	0.240	0.391	0.393	none	4.8
LEO L1	320,000	0.010	0.017	0.017	none	25.5

Table S5. Captured fraction at the ranges of the burns and relay links (0.05 µrad jitter).

S3. Mission studies

Fraction	5–95%	Median	Median Isp (s)	Closed
0.05	0.99–1.05	1.01	874	100%
0.10	0.98–1.09	1.02	891	100%
0.25	0.96–1.23	1.06	935	100%
0.50	0.95–1.47	1.13	998	100%
1.00	0.96–1.96	1.28	1,096	100%
2.00	1.01–3.20	1.62	1,256	100%

Table S6. IMLEO ratio against the useful laser power fraction (5–95%, median), median specific impulse and share of draws that close.

Receiver (kg/MW)	Design, median	Design, IQR	0.25, median	0.25, IQR
10	1.00	0.88–1.13	0.96	0.95–0.98
15	1.12	0.96–1.30	0.98	0.96–1.00
25	1.39	1.17–1.63	1.01	0.98–1.05
40	1.85	1.55–2.16	1.06	1.02–1.12
60	2.63	2.10–3.22	1.14	1.08–1.21
80	3.61	2.64–4.56	1.22	1.14–1.31

Table S7. IMLEO ratio against the receiver specific mass: median and interquartile range for the 4500 K design and for a laser fraction of 0.25.

Receiver (kg/MW)	ρ = 0.05	ρ = 0.1	ρ = 0.25	ρ = 0.5	ρ = 1	ρ = 1.5	ρ = 2	ρ = 2.4
2	0.99	0.98	0.96	0.94	0.91	0.91	0.90	0.91
4	0.99	0.98	0.97	0.96	0.95	0.96	0.96	0.97
6	0.99	0.99	0.98	0.98	0.99	1.00	1.02	1.04
8	0.99	0.99	0.99	1.00	1.03	1.05	1.08	1.12
10	1.00	1.00	1.00	1.02	1.06	1.11	1.15	1.20
15	1.00	1.01	1.03	1.07	1.15	1.23	1.32	1.40
20	1.01	1.02	1.05	1.12	1.24	1.37	1.50	1.60
30	1.02	1.04	1.11	1.22	1.43	1.64	1.86	2.03

Table S8. IMLEO ratio of the nominal vehicle on the grid of receiver specific mass and useful laser power fraction ρ.

Receiver (kg/MW)	ρ = 0.05	ρ = 0.1	ρ = 0.25	ρ = 0.5	ρ = 1	ρ = 1.5	ρ = 2	ρ = 2.4
2	385	415	538	685	907	1,198	1,499	1,831
4	459	571	766	974	1,797	3,321	4,119	5,571
6	631	838	1,050	1,936	13,581	none	none	none
8	921	1,217	2,523	32,841	none	none	none	none

Receiver (kg/MW)	ρ = 0.05	ρ = 0.1	ρ = 0.25	ρ = 0.5	ρ = 1	ρ = 1.5	ρ = 2	ρ = 2.4
10	1,073	3,792	none	none	none	none	none	none
15	none	none	none	none	none	none	none	none
20	none	none	none	none	none	none	none	none
30	none	none	none	none	none	none	none	none

Table S9. Number of missions over which the laser nodes must be amortized for the spacecraft saving to equal the node mass (none: LANTP is heavier than conventional NTP).

Time of flight (d)	Δv (km/s)	TLI burn (min)	IMLEO ratio
1.8	4.67	22	1.21
2.0	4.47	21	1.21
2.5	4.17	19	1.22
3.0	4.03	19	1.23
4.0	3.92	19	1.24
4.5	3.91	18	1.24

Table S10. Transfer time: total impulsive Δv (TLI + LOI, km/s), TLI burn time and IMLEO ratio of the nominal vehicle (receiver 10 kg/MW).

Payload (t)	NTP (t)	LANTP (t)	Ratio
5	21	38	1.86
10	31	46	1.49
20	53	64	1.21
40	101	100	1.00
80	231	174	0.75

Table S11. Payload: initial masses and ratio (receiver 10 kg/MW).

Parameter	Correlation
receiver specific mass	0.787
reactor power	0.558
chamber pressure	−0.137
nozzle recombination \phi	−0.076
tank fraction	−0.038
beam quality M^2	0.032
dropout rate	0.029
engine T/W	−0.022
re-acquisition time	0.022
pointing jitter	0.002

Table S12. Spearman rank correlation of the IMLEO ratio with the uncertain parameters.

Policy	5%	25%	50%	75%	95%	Closed
nominal (laser on both burns)	0.99	1.28	1.66	2.25	3.54	1.00
laser lost before TLI, reactor-only TLI (single fault)	1.40	1.77	2.28	3.18	4.66	1.00
reactor-only mission with the LANTP hardware (full fallback)	1.47	1.95	2.52	3.46	4.95	1.00

Table S13. Contingency policies: IMLEO ratio percentiles (5, 25, 50, 75, 95) and share of draws that close.

Fraction of burn	Propellant (t)	Burn time (min)
0.00	35.4	52
0.15	34.1	50
0.30	32.4	47
0.45	30.6	45
0.60	28.6	42
0.75	26.6	39
0.90	24.6	36
1.05	23.4	34

Fraction of burn	Propellant (t)	Burn time (min)
1.20	23.4	34

Table S14. TLI burn of the nominal vehicle after a loss of the laser at a fraction of the nominal burn time: propellant and burn time.

Payload (t)	Receiver (kg/MW)	NTP (t)	LANTP (t)	Ratio
10	10	59	76	1.28
10	25	59	140	2.36
20	10	no closure	97	–
20	25	no closure	164	–
40	10	820	149	0.18
40	25	820	761	0.93

Table S15. Round trip with propulsive return (four burns, reusable tug), nominal vehicle.

Output	5th	50th	95th
Isp at 4500 K design (s)	1,186.01	1,290.85	1,417.74
Effective Isp of TLI (s)	673.42	1,023.20	1,253.08
IMLEO ratio	1.02	1.69	3.70
IMLEO ratio with infrastructure	2.83	5.59	14.68
Emitted laser power (GW)	0.36	0.95	1.60
Electrical power at the node (GW)	0.89	2.34	4.31
Infrastructure mass (t)	8,281.66	19,789.63	54,805.26

Table S16. Percentiles of the main Monte Carlo outputs (1,500 draws).

S4. Reproducibility

The repository (https://github.com/sandlerleon/lantp-cislunar-mc, version 1.0.0) contains the model (lantp_core.py, lantp_mission.py), the studies (lantp_studies.py), the figure code (lantp_figures.py), the results (results/results.json), the automated tests (tests/test_all.py) and the manuscript sources. Commands: python lantp_studies.py --all runs every study (about 30 min on one core) and writes results/results.json and the figures; python tests/test_all.py runs the tests. Every study uses fixed seeds (base seed 2027; the Monte Carlo draws use seed 2029). Dependencies: Python 3, numpy, scipy, matplotlib; Cantera for the validation of the thermodynamics. The design inputs are the ranges of Table 2 (listed with their nominal values in results/results.json under params).